

Navigators Technical Documentation

Navigators — Contributor System & Workflow Specification

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1. Contributor Roles & Dual-Track Architecture

Navigators establishes two distinct contribution workflows: 1. **Local Contributor (Cartographic & Community Enrichment)**: Focuses on mapping road infrastructure, correcting geometry errors, and adding essential points of interest (POIs) such as fuel stations, EV chargers, and medical facilities. 2. **Internal Contributor (Telemetry & AI Model Engineering)**: Focuses on capturing calibrated vehicle sensor telemetry under controlled driving conditions to expand the training corpus for dead-reckoning models.

```
flowchart TD
    subgraph LocalTrack["Track 1: Local Contributor (Cartographic Track)"]
        User1["Public User"] -->|Drafts Place / POI| Contrib["Contribution Record (`status='draft'`")
        Contrib -->|Submits for Review| ReviewQueue["Moderator Review Queue (`status='pending_review'`")
        ReviewQueue -->|Moderator Evaluates| Decision{Approved?}
        Decision -->|Yes| CanonicalPlace["Canonical Map Place (`places` Table)"]
        Decision -->|No| RejectFeedback["Rejection with Notes (`status='rejected'`")
        CanonicalPlace -->|Online API Ingest| LiveMap["Live Map Rendering"]
        CanonicalPlace -->|Offline Sync Replay| LocalStorageCache["Client PWA Offline Cache"]
    end

    subgraph InternalTrack["Track 2: Internal Contributor (AI Engineering Track)"]
        User2["Authenticated User"] -->|Submits Access Application| AccessReq["Access Request (`in_progress`")
        AccessReq -->|Staff Review & Background Check| StaffReview{Approved?}
        StaffReview -->|Yes| UpgradeRole["Assign Role: `internal_contributor`"]
        UpgradeRole -->|Records Calibrated Drives| TelemetryIngest["Dataset Session Ingest (`dataset_id`")
        TelemetryIngest -->|Automated Data Quality Audit| DataValidation{Continuity & No-NaN?}
        DataValidation -->|Yes| CuratedDataset["Curated Training Split (`datasets` Table)"]
        CuratedDataset -->|Launches PyTorch Training| CandidateModel["Candidate Model (`models` Table)"]
    end
```

2. Track 1: Local Contributor Lifecycle

2.1 Contribution State Machine

All crowdsourced cartographic additions execute the deterministic lifecycle governed by `ContributionStateMachine` in `src/db/state_machine.py` : - `DRAFT` : Author creates and edits place fields privately. Invisible to non-staff users. - `SUBMITTED` : Author locks edits and queues the contribution for moderation. - `PENDING_REVIEW` : Automated triage flags category, spatial bounds, and duplicates. - `APPROVED` : An independent Moderator (not the author) inspects coordinates and approves the entry. - `PUBLISHED` : System transactionally creates or updates the corresponding entity in the canonical `places` table and updates `target_resource_id`. - `CHANGES_REQUESTED` : Reviewer asks for clarifications; contribution returns to draft status for author updates. - `REJECTED` : Submission fails quality criteria and is closed with explanatory notes. - `WITHDRAWN` : Author cancels submission prior to final moderation decision.

2.2 Ownership & IDOR Protection

- Authors possess exclusive modification rights on their own drafts (`action='contribution:update'`).
 - Third-party users cannot view, edit, or delete another user's drafts.
 - Moderators cannot review their own submissions (self-moderation is blocked by `AuthorizationService`).
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3. Track 2: Internal Contributor Lifecycle

3.1 Access Request Protocol

Because raw sensor telemetry collection and neural training require trusted data provenance, the `internal_contributor` role is not granted automatically: 1. An authenticated user submits `POST /api/v1/internal-contributors/request` specifying hardware setup, vehicle type, and experience. 2. A record is created in `internal_access_requests` with `status = 'pending'`. 3. A Team Admin or Super Admin inspects the application. Upon approval, the server assigns `internal_contributor` role in `user_roles`.

3.2 Telemetry Ingestion & Consent Verification

1. Internal contributors upload recorded driving sessions via `POST /api/v1/datasets/sessions`.
 2. The payload must include `consent: true`. Ingest without explicit consent is rejected.
 3. The dataset validation worker verifies:
 - Zero negative time steps ($\Delta t > 0$).
 - Sensor gap threshold ≤ 1.0 s.
 - Realistic acceleration bounds ($\|a\| \leq 40 \text{ m/s}^2$).
 - Valid GNSS fix coordinates for training ground truth.
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